BACKUP KIT · SPEC SHEET OCT 21, 2026

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48 V $LiFePO_4$ lithium · 10 h

RUNNING LOAD	START-UP PEAK	BATTERY NEEDED	RUNTIME
465 W	**1,515 W**	**4.8 kWh**	**~20 h**
~368 W average			18.1 h on a hard day · 21.3 h on a light day

Parts list

Inverter	3,000 W continuous, not VA, pure sine, 48 V DC input 6,000 W surge · covers 465 W running, 1,515 W peak	**×1**
Batteries	48 V 200 Ah $LiFePO_4$ 1 in series × 1 in parallel (1S1P) · 9.6 kWh bank, 8.2 kWh usable · 200 A continuous, 400 A peak from the bank (assumed ratings)	**×1**
Main cable	2 AWG (33.6 mm²) copper, 1.5 m each Battery to inverter · 87 A design current · 115 A after derating · 0.2% drop, 0.1% at start-up	**×2**
Fuse	100 A class T Inline on the positive lead, as close to the bank as possible · 20 kA interrupting rating	**×1**

An estimate for requesting quotes, not an installation design. Have a licensed electrician verify it against the actual equipment and your local electrical code before you buy or install anything.

Every appliance counts in every hour of the window. Assumes 90% inverter efficiency, 25% inverter headroom, 85% depth of discharge, 1,100 W per ton of cooling, one motor starting at a time, copper cable with 90 °C insulation at 30 °C. NEC ampacity tables; verify against local code. Inverter sizes are continuous watts, not VA, at room temperature; the 25% headroom covers power factor and heat. Assumed ratings, not a maker's datasheet: each 48 V 200 Ah $LiFePO_4$ battery gives 200 A continuous and 400 A peak for 5 s, at most 16 in parallel, and 25 A per Ah into a short; each 3,000 W inverter surges to 6,000 W for 5 s.

Load schedule

APPLIANCE	QUANTITY	EACH	RUNNING LOAD	AVERAGE	OVER 10 H
ALWAYS ON					
LED lighting (10 bulbs)	1	90 W	90 W	90 W	900 Wh
TV, 55-inch LED	1	120 W	120 W	120 W	1,200 Wh
Wi-Fi router + modem	1	20 W	20 W	20 W	200 Wh
Laptop	1	65 W	65 W	65 W	650 Wh
Phone charger	2	10 W	20 W	20 W	200 Wh
CYCLES ON AND OFF					
Refrigerator	1	150 W	150 W	53 W	525 Wh
Total	**7**		**465 W**	**368 W**	**3,675 Wh**

Checks

- **Inverter size**
  465 W running on a 3,000 W inverter · 1,515 W for 3 s, within its assumed 6,000 W surge rating
  465 W running + 1,050 W (Refrigerator) starting
- **Battery continuous current**
  Bank gives an assumed 200 A continuous; the inverter draws up to 69 A
- **Battery peak current**
  Bank peak of an assumed 400 A covers the 35 A start-up for 3 s
- **Parallel strings**
  One string, nothing to balance · assumed limit of 16 in parallel
- **Fuse**
  100 A class T: under the cable's 115 A, 20 kA interrupting, start-up 35 A within what it holds
- **Cable**
  2 AWG carries 115 A after derating (30 °C) for a 87 A design current · 0.2% drop, 0.1% at start-up